

## 8.012 Problem Set 5

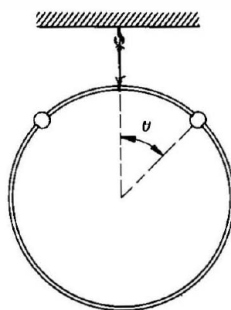
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### Problem 3

Kleppner & Kolenkow, Problem 4.7

4.7 A ring of mass  $M$  hangs from a thread, and two beads of mass  $m$  slide on it without friction (see above right). The beads are released simultaneously from the top of the ring and slide down opposite sides. Show that the ring will start to rise if  $m > 3M/2$ , and find the angle at which this occurs. *Ans. clue.* If  $M = 0$ ,  $\theta = \arccos \frac{2}{3}$



Forces acting on each Bead :-

1. Normal Force
2. Gravity

(The centripetal force is not an additional force; it is provided by the radial components of the real forces.)

The Normal Force between a Bead and the Ring acts along the radius of the ring passing through the bead and is directed radially outward.

Taking the radial direction toward the center of the ring as positive, the radial force balance for each bead is given by

$$mg \cos(\theta) - N = \frac{mv^2}{R}$$

$$\implies N = mg \cos(\theta) - \frac{mv^2}{R}$$

The vertical component of the force exerted by one bead on the ring is downward and has magnitude  $N \cos(\theta)$ . Hence, the total downward force exerted by the two beads on the ring is

$$\begin{aligned} 2N \cos(\theta) &= 2 \left( mg \cos(\theta) - \frac{mv^2}{R} \right) \cos(\theta) \\ &= 2mg \cos^2(\theta) - \frac{2mv^2 \cos(\theta)}{R} \end{aligned}$$

The ring begins to rise at the instant when the downward force exerted by the two beads equals the weight of the ring; beyond this point the net vertical force on the ring is upward.

Thus, the condition for the ring to start rising is

$$2N \cos(\theta) \leq Mg$$

Substituting for  $N$ ,

$$2mg \cos^2(\theta) - \frac{2mv^2 \cos(\theta)}{R} \leq Mg \dots (i)$$

Using Energy Conservation

$$v^2 = 2gR(1 - \cos(\theta))$$

In-equation (i) becomes

$$\begin{aligned} 2mg \cos^2(\theta) - 4mg \cos(\theta)(1 - \cos(\theta)) &\leq Mg \\ \implies 6m \cos^2(\theta) - 4m \cos(\theta) &\leq M \end{aligned}$$

This in-equation determines the angle  $\theta$  at which the ring begins to rise. The equality corresponds to the instant the ring just loses contact with the thread; the inequality determines whether such an angle exists.

Now  $6m \cos^2(\theta) - 4m \cos(\theta)$  is Maximum at

$$\cos(\theta) = \frac{4m}{12m} = \frac{1}{3}$$

So

$$[6m \cos^2(\theta) - 4m \cos(\theta)]_{MAX} = 6m \frac{1}{9} - 4m \frac{1}{3} = -\frac{2m}{3}$$

This leads the Inequality to convert into

$$-\frac{2m}{3} \leq M \implies m \geq \frac{3M}{2}$$

## Problem 5

Kleppner & Kolenkow, Problem 4.9

**4.9** A simple and very violent chemical reaction is  $\text{H} + \text{H} \rightarrow \text{H}_2 + 5 \text{ eV}$ . (1 eV =  $1.6 \times 10^{-19}$  J, a healthy amount of energy on the atomic scale.) However, when hydrogen atoms collide in free space they simply bounce apart! The reason is that it is impossible to satisfy the laws of conservation of momentum and conservation of energy in a simple two body collision which releases energy. Can you prove this? You might start by writing the statements of conservation of momentum and energy. (Be sure to include the energy of reaction in the energy equation, and get the sign right.) By eliminating the final momentum of the molecule from the pair of equations, you should be able to show that the initial momenta would have to satisfy an impossible condition.

Let each hydrogen atom have mass  $m$ . Let the initial momenta of the two atoms be  $\vec{p}_1$  and  $\vec{p}_2$ . After the reaction, the hydrogen molecule  $\text{H}_2$  has mass  $2m$  and momentum  $\vec{P}$ .

### Conservation of Momentum

Since no external forces act,

$$\vec{p}_1 + \vec{p}_2 = \vec{P}. \quad (1)$$

### Conservation of Energy

The initial kinetic energy is

$$K_i = \frac{p_1^2}{2m} + \frac{p_2^2}{2m}.$$

The final kinetic energy of the molecule is

$$K_f = \frac{P^2}{2(2m)} = \frac{P^2}{4m}.$$

Since energy  $Q$  is released,

$$K_f = K_i + Q. \quad (2)$$

### Elimination of the Final Momentum

From momentum conservation (1),

$$P^2 = (\vec{p}_1 + \vec{p}_2)^2 = p_1^2 + p_2^2 + 2\vec{p}_1 \cdot \vec{p}_2.$$

Substituting into (2),

$$\frac{p_1^2 + p_2^2 + 2\vec{p}_1 \cdot \vec{p}_2}{4m} = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} + Q.$$

$$\implies p_1^2 + p_2^2 + 2\vec{p}_1 \cdot \vec{p}_2 = 2p_1^2 + 2p_2^2 + 4mQ.$$

$$\implies 2\vec{p}_1 \cdot \vec{p}_2 = p_1^2 + p_2^2 + 4mQ. \quad (3)$$

### Impossibility of the Condition

By the Cauchy–Schwarz inequality,

$$2\vec{p}_1 \cdot \vec{p}_2 \leq 2|\vec{p}_1||\vec{p}_2| \leq p_1^2 + p_2^2.$$

However, equation (3) requires

$$2\vec{p}_1 \cdot \vec{p}_2 = p_1^2 + p_2^2 + 4mQ > p_1^2 + p_2^2,$$

since  $Q > 0$ .

This is a contradiction.

Hence,

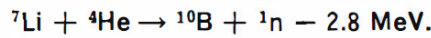
**A two–body collision in free space cannot simultaneously satisfy conservation of momentum and conservation of energy if energy is released.**

**Therefore, the reaction  $\text{H} + \text{H} \rightarrow \text{H}_2 + 5\text{ eV}$  cannot occur in free space.**

## Problem 9

Kleppner & Kolenkow, Problem 4.28

4.28 A thin target of lithium is bombarded by helium nuclei of energy  $E_0$ . The lithium nuclei are initially at rest in the target but are essentially unbound. When a helium nucleus enters a lithium nucleus, a nuclear reaction can occur in which the compound nucleus splits apart into a boron nucleus and a neutron. The collision is inelastic, and the final kinetic energy is less than  $E_0$  by 2.8 MeV. (1 MeV =  $10^6$  eV =  $1.6 \times 10^{-13}$  J). The relative masses of the particles are: helium, mass 4; lithium, mass 7; boron, mass 10; neutron, mass 1. The reaction can be symbolized

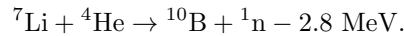


a. What is  $E_{0,\text{threshold}}$ , the minimum value of  $E_0$  for which neutrons can be produced? What is the energy of the neutrons at this threshold?

Ans. Neutron energy = 0.15 MeV

b. Show that if the incident energy falls in the range  $E_{0,\text{threshold}} < E_0 < E_{0,\text{threshold}} + 0.27 \text{ MeV}$ , the neutrons ejected in the forward direction do not all have the same energy but must have either one or the other of two possible energies. (You can understand the origin of the two groups by looking at the reaction in the center of mass system.)

The nuclear reaction is



The lithium nucleus is initially at rest in the laboratory frame, while the helium nucleus has initial kinetic energy  $E_0$ .

The relative masses (in atomic mass units) are:

$$m_{\text{He}} = 4, \quad m_{\text{Li}} = 7, \quad m_{\text{B}} = 10, \quad m_{\text{n}} = 1.$$

a.

At threshold, the reaction products move together with the same velocity in the center-of-mass frame. Hence, the final kinetic energy in the center-of-mass frame is zero.

**Momentum conservation (laboratory frame):**

$$p_{\text{initial}} = p_{\text{final}}.$$

Let the initial momentum of the helium nucleus be  $p$ . Then,

$$p = (m_{\text{B}} + m_{\text{n}})V.$$

**Energy conservation:**

$$E_0 = \frac{p^2}{2m_{\text{He}}}$$

$$E_{\text{final}} = \frac{p^2}{2(m_{\text{B}} + m_{\text{n}})}.$$

Including the reaction energy loss,

$$\frac{p^2}{2m_{\text{He}}} = \frac{p^2}{2(m_{\text{B}} + m_{\text{n}})} + 2.8 \text{ MeV}.$$

Solving for  $p^2$ ,

$$p^2 = \frac{2(2.8 \text{ MeV})}{\frac{1}{m_{\text{He}}} - \frac{1}{m_{\text{B}} + m_{\text{n}}}}.$$

Substituting numerical values,

$$p^2 = 2(2.8) \frac{(4)(11)}{11 - 4} = 2(2.8) \frac{44}{7}.$$

Thus the threshold energy is

$$E_{0,\text{threshold}} = \frac{p^2}{2m_{\text{He}}} = 2.8 \frac{m_{\text{B}} + m_{\text{n}}}{m_{\text{B}} + m_{\text{n}} - m_{\text{He}}} = 2.8 \frac{11}{7} \approx 4.4 \text{ MeV}.$$

$$E_{0,\text{threshold}} \approx 4.4 \text{ MeV}$$

At threshold, the neutron and boron nucleus move together, so the neutron kinetic energy is

$$E_n = \frac{m_{\text{n}}}{m_{\text{B}} + m_{\text{n}}} (E_{0,\text{threshold}} - 2.8).$$

$$E_n = \frac{1}{11} (4.4 - 2.8) \approx 0.15 \text{ MeV}.$$

$$E_n = 0.15 \text{ MeV}$$

**b.**

For

$$E_{0,\text{threshold}} < E_0 < E_{0,\text{threshold}} + 0.27 \text{ MeV},$$

the reaction products have nonzero kinetic energy in the center-of-mass frame.

In the center-of-mass frame:

- The neutron and boron nucleus recoil with equal and opposite momenta.
- Their kinetic energies are fixed by energy conservation.

Transforming back to the laboratory frame, the neutron velocity is obtained by vector addition of:

1. The center-of-mass velocity
2. The neutron velocity in the center-of-mass frame

Since the neutron may be emitted either forward or backward in the center-of-mass frame, its laboratory-frame velocity (and hence kinetic energy) can take **two distinct values**.

**For  $E_{0,\text{threshold}} < E_0 < E_{0,\text{threshold}} + 0.27 \text{ MeV}$ ,  
forward neutrons occur in two distinct energy groups.**